

Juan David Muñoz-Bolaños

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juan.munoz@i-med.ac.at

MB Automation GmbH & Co. KG

Frau Tugce Zabun
Mühlbauer Group
Roding
Deutschland

Subject: Bewerbung als Application Engineer Image Processing (m/w/d)

Sehr geehrte Frau Zabun,

als Physikingenieur mit Fokus auf industrieller Bildverarbeitung bewerbe ich mich für die Position in Roding. Bei INTECOL S.A.S. habe ich bereits Vision-Lösungen (Halcon/Cognex) für Kunden wie Renault realisiert, wobei die präzise Abstimmung einzelner Optik- und Beleuchtungskomponenten entscheidend war.

In meiner Promotion habe ich komplexe Mikroskopsysteme aufgebaut und hochpräzise optomechanische Komponenten (Thorlabs, Edmund Optics) qualifiziert. Diese Hands-on-Erfahrung in der Justage und Hardware-Software-Integration ermöglicht es mir, Vision-Komponenten zielsicher zu optimieren und QA-Prozesse in der Montage zu sichern.

Bestens in die DACH-Arbeitskultur integriert (B2), stehe ich Ihnen gerne auch für internationale Serviceeinsätze zur Verfügung. Ich freue mich darauf, meine Praxiserfahrung und Begeisterung für Optik in Ihr Team einzubringen.

Mit freundlichen Grüßen,

Juan David Muñoz-Bolaños

Application Engineer Image Processing
Medizinische Universität Innsbruck

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PROFESSIONAL SUMMARY

Experienced Physics Engineer specialized in industrial machine vision, optical design, and hardware-software integration. Proven background in commissioning custom vision solutions on production lines. Expert in choosing the right illumination, cameras, and lenses to ensure high-performance quality assurance in automated manufacturing environments.

TECHNICAL COMPETENCIES

Machine Vision Hardware

Mastering of CMOS/CCD camera integration, lens selection (telecentric, macro), and custom illumination design (Dark-field, Bright-field, coaxial).

Image Processing Libraries

Advanced proficiency in Halcon and Cognex VisionPro for real-time inspection, part recognition, and measurement. Familiar with OpenCV and Python-based vision pipelines.

System Alignment & Justage

Hands-on experience in the precise alignment of complex opto-mechanical setups (Laser sources, scanners, detectors). Expert in qualifying components from market-leading vendors.

Automation & Integration

Skilled in automating optical systems with Python, C++, and LabVIEW. Experienced in ensuring Gage R&R and measurement reliability in harsh industrial environments.

R&D Methodology

Experience bringing systems from concept to series prototypes, including ZEMAX modeling, procurement management, and technical documentation.

PROFESSIONAL EXPERIENCE

INTECOL S.A.S.

Junior Machine Vision System Engineer

Medellín, Colombia

Mar 2018 – Mar 2019

- **Vision Pipeline Engineering:** Designed and implemented Halcon-based beverage bottle inspection systems; developed custom illumination strategies to reliably detect liquid levels and printing defects.
- **Industrial Integration:** Commissioned Cognex VisionPro systems for auto-part recognition at Renault; integrated vision sensors with robotic arms for flame-surface treatment processes.
- **QA & Feasibility:** Performed systematic Gage R&R assessments and feasibility studies for new industrial clients, ensuring 24/7 reliability of vision components on the floor.

Medical University of Innsbruck

PhD Researcher – Optical Metrology & Automation

Innsbruck, Austria

Jan 2022 – Present

- **System Build & Justage:** Led the end-to-end assembly of a custom multiphoton microscope; adjusted galvo scanners, laser sources, and PMTs to achieved sub-micron resolution.
- **Adaptive Optics Control:** Programmed a Python/C++ control stack to synchronize scientific cameras with active wavefront correctors for real-time sample characterization.
- **Vendor & Component QM:** Managed technical specification sheets and qualified precision hardware from Thorlabs and Edmund Optics as part of system procurement.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher (Photonics & Data)

Jena, Germany

Jun 2020 – Dec 2021

- **Instrumentation Design:** Prototyping of a dispersed fiber-probe Raman spectrometer; handled optical layout justage and hardware-software calibration.

- **Data Evaluation Hooks:** Integrated computational data analysis (Python ML) with hardware sensors to optimize measurement throughput and validation.

EDUCATION

Medical University of Innsbruck
Ph.D. in Adaptive Optics & Microscopy

Austria
Jan 2022 – Present

Friedrich Schiller University Jena
M.Sc. in Photonics

Germany
2019 – 2021

University of Cauca
B.Sc. in Physics Engineering

Colombia
2012 – 2018

PUBLICATIONS & AWARDS

- **Awards:** ASML & ZEISS Summer Schools participant; COLFUTURO Grant 2026 awardee.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte
Head of Institute
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft
Spectroscopy Group Leader
christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy





URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

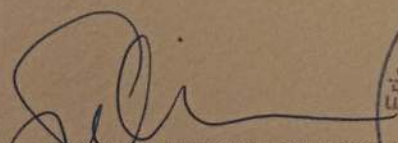
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



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österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

